Contact Center Compliance API Best Practices

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Below is a collection of best practices based on common questions and issues we see in API integration.

1 Use TLS 1.2 or above

All API requests to DNCScrub must be made with HTTPS and TLS 1.2.

2 Pass clean data to the API

All phone numbers should be formatted as 10 consecutive digits example: 5039367187. Phone numbers should not contain alphabetical characters, a leading 1, an extension, or formatting characters.

Although our system does intelligently strip out formatting characters (see FAQ below), it's been our observation that when non-clean data is passed to the API (such as just passing a free form field with no validation), there can be unexpected issues in the software calling the API.

3 API request rate limit is 100 requests per minutes per account

Every DNCScrub account is limited to making 100 API requests per minute. Each request can contain up to 10,000 numbers. We recommend submitting multiple phone numbers per request when possible vs lots of individual requests.

4 If you submit multiple phone number per request understand the API is atomic

Be default the API is atomic. When multiple phone numbers are submitted either 1) Action is taken for each phone number, or 2) Action is not taken on any of the phone numbers. For example if you pass an IDNC Add request and one of the phone numbers is invalid, none of the phone numbers submitted are inserted into your IDNC database (an error is returned stating this). Likewise if you perform a scrub and one of the phone numbers is invalid, for example "AAA", none of your phone numbers will be scrubbed.

5 If scrubbing more than 50 numbers use HTTP POST

Our system will not accept large HTTP GET requests (large HTTP GET requests violate the HTTP specification). If you are scrubbing more than 50 numbers, you must use HTTP POST.

6 Consider how you wish to handle phone numbers outside the US

Phone numbers within the North America Numbering Plan but outside the US include Canada phone numbers as well as phone numbers in Bahamas, Anguilla, British Virgin Islands, and others. Consider how you want your system to respond to these phone numbers and include phone numbers outside the US in your test cases.

7 Use an API appropriate timeout and exponential back-off on retry

The API has high reliable with multi server redundancy. However best practice is to still implement appropriate timeouts and retry mechanism in case of exception circumstances.

7.1 Timeout value

If you pass less than 200 records we recommend doing 4 seconds times the number of records.

On 1 record the timeout would be 4 seconds
3 records 12 second timeout
5 records 20 second timeout
10 records 40 second timeout

If you pass more than 200 records we recommend a 600 second time out.

7.2 Retry intervals

If no response is received within the timeout, the request should be resubmitted waiting 4 seconds between requests and using an exponential back-off. We recommend retrying 5 times, exponentially take a longer time each request, and after the 5th retry notifying a system operator.

Frequently Asked Question

Q: What is the maximum amount of numbers that can be passed at one time?

A: With HTTP GET you can pass up to 125 numbers. With HTTP POST you can pass up to 10,000. We do not recommend batches larger than 10,000 records.

Q: Can phone numbers in the phoneList parameter contain formatting characters such as (232) 123-1234?

A: Yes. The API determines the phone number as the first 10 consecutive digits in the phoneList value. So if **(232) 123-1234** it determines the phone number as 2321231234. Likewise is ABC232 123 123AA4 is passed it would read the phone number as 2321231234. Note that if the phoneList field contains more than 10 digits, the API assume the first 10 digits are the phone number. Example if you pass in 2322121234x100 it would scrub 2322121234

Note that although we strip out formatting characters we do not recommend they are in requests as they lead to confusion debugging.

Q: Can phone numbers in the phoneList parameter contain a leading 1 that the API ignores?

A: **No.** The API must only be passed in valid phone numbers in the North America Numbering Plan in 10 digit format. A phone number passed in such 1 232 123 1234 will not return a valid result. Instead 232 123 1234 should be passed in The API assumes the first 10 digits in the phoneList parameter are what are to scrubbed so if for example you pass in 12321231234 it determines the phone number as 123 212 3123.

Q: How are values in the phoneList parameter handled that are not proper phone numbers as they contain less than 10 digits. For example what happens if we pass in a phone number of AAA BBB CCCC

A: An HTTP 400 response will be returned with human readable message indicating the phone number passed is not valid.